

EXECUTIVE SUMMARY

OncoGraphRAG: Biomedical GraphRAG for Breast Cancer Literature Mining

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15 Key Points for Management Review

Context / Strategy

Completed / Positive

Attention / Next Steps

01

WHAT WE BUILT

A working AI system that answers breast cancer research questions from PubMed literature

OncoGraphRAG is a complete, locally running prototype that lets a researcher ask a medical question in plain English and receive an answer drawn from real published literature — with source citations traceable back to PubMed articles.

02

THE PROBLEM IT SOLVES

Researchers can't read thousands of papers — we extract the knowledge automatically

PubMed has over 36 million abstracts. For any given breast cancer topic (e.g. nanoparticle drug delivery), a researcher would need to manually sift through hundreds of papers. Our system retrieves, organises, and summarises the relevant evidence in seconds.

03

SCALE OF WORK

28,638 medical articles indexed; 7,695 structured relationships extracted

The system has ingested and processed a large PubMed oncology corpus. Every article chunk is searchable by meaning (not just keywords), and 7,695 typed biomedical relationships — e.g. 'doxorubicin delivered via nanoparticles' — have been automatically extracted and stored in a knowledge graph.

04

TWO-LAYER RETRIEVAL

The system finds answers two ways: by meaning AND by medical relationships

Most AI search tools use only semantic similarity. OncoGraphRAG adds a second layer — a knowledge graph — that captures typed medical relationships (e.g. DRUG—TREATS—DISEASE). This reduces the risk of surfacing articles that sound similar but describe different mechanisms.

05

KNOWLEDGE GRAPH

10 types of biomedical relationships encoded — TREATS, TARGETS, DELIVERED_VIA, and more

The system automatically labels relationships between medical entities: drugs, diseases, genes, biomarkers, therapies, assays, cell types, and biological mechanisms. For example: 'trastuzumab —TARGETS→ HER2' or 'Oncotype DX —MEASURES→ recurrence score'.

06

AI ANSWER GENERATION**A language model reads the retrieved evidence and writes a coherent answer**

After retrieval, the Gemma instruction-tuned language model synthesises a narrative answer from the evidence — similar to how a medical researcher would write a literature summary. The answer is grounded in retrieved PubMed passages, not generated from the model's memory alone.

07

EVIDENCE TRACEABILITY**Every answer is linked back to its PubMed source — no black-box outputs**

Each retrieved passage carries a PubMed ID (PMID) and URL. Each relationship triple preserves the verbatim sentence it was extracted from and the originating article. This means any claim in the answer can be independently verified against the source publication.

08

RUNS LOCALLY**The entire system runs on a standard laptop GPU — no cloud API required**

The prototype operates on an NVIDIA RTX 3050 laptop GPU (4GB VRAM) with no dependency on paid cloud services. This is important for data security and for demonstrating that the architecture is cost-effective and feasible to deploy in research or clinical environments.

09

LIVE TEST RESULT**Test query on nanoparticle drug delivery retrieved accurate, relevant evidence**

The query 'How are nanoparticles used in breast cancer drug delivery?' successfully retrieved PubMed evidence on precision nanotechnology and drug-resistant breast cancer, generated a biologically accurate answer covering efflux pump circumvention, immune modulation, and controlled drug release.

10

SAFETY BUILT IN**The system does not generate clinical advice or patient-level recommendations**

The prototype is strictly a literature mining tool. It does not provide clinical diagnoses, treatment recommendations, or dosing instructions. All outputs are intended for research synthesis review by qualified biomedical experts — not direct clinical use.

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KNOWN LIMITATIONS**Citation accuracy and relationship specificity need improvement before research-grade use**

The Gemma 1B model does not consistently attach PubMed citations to every generated claim. A portion of extracted relationships use a generic fallback label ('ASSOCIATED_WITH') rather than a specific biological relation type. Both are targeted in the next development phase.

12

EVALUATION PLAN**Formal evaluation planned: 100 expert-curated questions across 7 breast cancer topics**

A structured benchmark of 100 questions spanning drug delivery, therapy resistance, cancer

stem cells, biomarkers, assays, molecular mechanisms, and combination therapy has been designed. Metrics include retrieval recall, citation accuracy, answer faithfulness, hallucination rate, and expert usefulness score.

13**UPGRADE ROADMAP****Five planned upgrades: better NER, relation classifiers, hybrid search, graph DB, larger models**

Next steps include: (1) replacing dictionary NER with BioBERT/scispaCy; (2) adding a supervised relation extraction classifier; (3) combining keyword and semantic search; (4) migrating to a Neo4j graph database for multi-hop queries; (5) evaluating larger biomedical models such as MedGemma and BioMistral.

14**REPRODUCIBILITY****All pipeline stages produce inspectable output files — fully reproducible**

The system is built as 8 numbered Python scripts, each reading and writing explicit files (JSONL, CSV, FAISS binary). Every stage can be re-run, inspected, or replaced independently. This modular design makes the system auditable, debuggable, and extensible by any team member.

15**STRATEGIC VALUE****Foundation for a clinical-grade biomedical evidence retrieval product**

OncoGraphRAG demonstrates the technical feasibility of automated, evidence-grounded biomedical question answering over real literature. With the planned upgrades, this system is a viable foundation for applications in oncology literature review, drug repurposing research, and clinical knowledge translation tools.

This document summarises the OncoGraphRAG manuscript draft (2026). For full technical detail, methodology, and evaluation design refer to the companion JBI manuscript.